535510 Reinforcement Learning

Pre-Lecture Assignment 1

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Policy Gradient Implemented in Python

In Lectures 4-6, we will learn various different PG algorithms. Let's try to match the concept of PG with the actual Python implementation. You can refer to the slides of Lecture 4 as a starting point when answering the questions below.

1) PG expression: The **PG.py** is a Python file generated by GPT. This piece of code implements one of the PG expressions. Take a read through the code. Can you figure out which PG expression does ChatGPT implement in **PG.py** (*i.e.*, either (P1), (P2), or (P3))?

2) A deeper look at the code:

(a) What kind of policy parameterization is GPT using here?

(b) In Line 35, why does GPT use the "reversed" operation?

(c) What is Line 41 for?

3) Do you fully agree with GPT?: Can you either identify 1 bug in the code or provide at least 1 suggestion for improving the code?

▸ **Expressions of Policy Gradient (aka *Policy Gradient Theorem*):**

(P1) Total reward: $$\nabla_\theta V^{\pi_\theta}(\mu) = \mathbb{E}_{\tau \sim P_\mu^{\pi_\theta}}\Big[G(\tau) \sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\Big]$$

(P2) REINFORCE: $$\nabla_\theta V^{\pi_\theta}(\mu) = \mathbb{E}_{\tau \sim P_\mu^{\pi_\theta}}\Big[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_\theta}(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t \mid s_t)\Big]$$

(P3) Q-value and discounted state visitation:

$$\nabla_\theta V^{\pi_\theta}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot \mid s)}\Big[Q^{\pi_\theta}(s, a) \nabla_\theta \log \pi_\theta(a \mid s)\Big]$$

Submission Guidelines and Remarks

- Please write all your responses within **1 page** (**A4 paper**, either hand-written or typeset). The responses that go beyond 1 page will *not* be graded.

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.

- Please submit it by **9pm, 3/17 (Tuesday)**.